

FILTER DESIGN TEST CASE

Low-Pass Filter - Band 28 (700 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Low-Pass Filter filter targeting Band 28 (700 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on Langasite substrate with a target center frequency of 4361.3 MHz and bandwidth of 235.0 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Low-Pass Filter
- Frequency Band: Band 28 (700 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: Langasite
- Package: Flip-Chip BGA
- Design Tool: Cadence AWR Microwave Office
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Low-Pass Filter design targeting Band 28 (700 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	4361.3 MHz
Bandwidth (3 dB):	235.0 MHz
Insertion Loss Target:	≤ 2.9 dB
Return Loss Target:	≥ 8.0 dB
Out-of-Band Rejection:	≥ 30.0 dB
Isolation (Duplexer):	≥ 27.0 dB
Q Factor:	6888
Temperature Coefficient:	-25.5 ppm/°C
Die Size:	2.1 x 1.7 mm
Wafer Size:	8-inch
Number of Resonators:	6
Electrode Material:	Mo

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the Low-Pass Filter template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 4361.3 MHz, BW 235.0 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.9 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, Langasite).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 13084 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.9 dB
- Return loss in passband shall be >= 8.0 dB
- Out-of-band rejection at +/- 352 MHz offset >= 30.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 13.9 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 95%

6. TEST RESULTS

Measurement Date: 2024-07-07
Devices Tested: 38
Insertion Loss (meas): 2.59 dB
Return Loss (meas): 10.1 dB
Rejection (meas): 33.1 dB
Isolation (meas): 32.4 dB
Q Factor (meas): 6888
Group Delay Variation: 1.2 ns
Power Handling: +34 dBm
Die Yield: 91.2%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Low-Pass Filter design demonstrated acceptable performance for Band 28 (700 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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